

ATT&CKcon 7.0 in-person tickets are open! Join us October 27-28, 2026 in McLean, VA. [Register here](#) for in-person tickets; hotel and location details can be found in the [FAQ](#).

[Home](#) > [Techniques](#) > [Enterprise](#) > [Phishing](#) > Spearphishing Attachment

Phishing: Spearphishing Attachment

Other sub-techniques of Phishing (4)

Adversaries may send spearphishing emails with a malicious attachment in an attempt to gain access to victim systems. Spearphishing attachment is a specific variant of spearphishing. Spearphishing attachment is different from other forms of spearphishing in that it employs the use of malware attached to an email. All forms of spearphishing are electronically delivered social engineering targeted at a specific individual, company, or industry. In this scenario, adversaries attach a file to the spearphishing email and usually rely upon [User Execution](#) to gain execution.^[1] Spearphishing may also involve social engineering techniques, such as posing as a trusted source.

There are many options for the attachment such as Microsoft Office documents, executables, PDFs, or archived files. Upon opening the attachment (and potentially clicking past protections), the adversary's payload exploits a vulnerability or directly executes on the user's system. The text of the spearphishing email usually tries to give a plausible reason why the file should be opened, and may explain how to bypass system protections in order to do so. The email may also contain instructions on how to decrypt an attachment, such as a zip file password, in order to evade email boundary defenses. Adversaries frequently manipulate file extensions and icons in order to make attached executables appear to be document files, or files exploiting one application appear to be a file for a different one.

ID: T1566.001

Sub-technique of: [T1566](#)

Tactic: [Initial Access](#)

Platforms: Linux, Windows, macOS

Contributors: Philip Winther

Version: 2.2

Created: 02 March 2020

Last Modified: 12 May 2026

ⓘ

ⓘ

[Version Permalink](#)

Procedure Examples

ID	Name	Description
C0028	2015 Ukraine Electric Power Attack	During the 2015 Ukraine Electric Power Attack , Sandworm Team obtained their initial foothold into many IT systems using Microsoft Office attachments delivered through phishing emails. ^[2]
G0018	admin@338	admin@338 has sent emails with malicious Microsoft Office documents attached. ^[3]
S0331	Agent Tesla	The primary delivered mechanism for Agent Tesla is through email phishing messages. ^[4]
G0130	Ajax Security Team	Ajax Security Team has used personalized spearphishing attachments. ^[5]
G0138	Andariel	Andariel has conducted spearphishing campaigns that included malicious Word or Excel attachments. ^{[6][7]}
S0622	AppleSeed	AppleSeed has been distributed to victims through malicious e-mail attachments. ^[8]
G0099	APT-C-36	APT-C-36 has used spearphishing emails with malicious .pdf and .docx files and password protected RAR attachments to avoid being detected by the email gateway. ^{[9][10][11][11]}
G0006	APT1	APT1 has sent spearphishing emails containing malicious attachments. ^[12]
G0005	APT12	APT12 has sent emails with malicious Microsoft Office documents and PDFs attached. ^{[13][14]}
G0073	APT19	APT19 sent spearphishing emails with malicious attachments in RTF and XLSM formats to deliver initial exploits. ^[15]
G0007	APT28	APT28 sent spearphishing emails containing malicious Microsoft Office and RAR attachments. ^{[16][17][18][19][20][21][22][23][24]}
G0016	APT29	APT29 has used spearphishing emails with an attachment to deliver files with exploits to initial victims. ^{[25][26][27][28]}
G0013	APT30	APT30 has used spearphishing emails with malicious DOC attachments. ^[29]
G0050	APT32	APT32 has sent spearphishing emails with a malicious executable disguised as a document or spreadsheet. ^{[30][31][32][33][34][35]}
G0064	APT33	APT33 has sent spearphishing e-mails with archive attachments. ^[36]
G0067	APT37	APT37 delivers malware using spearphishing emails with malicious HWP attachments. ^{[37][38][39]}
G0082	APT38	APT38 has conducted spearphishing campaigns using malicious email attachments. ^[40]
G0087	APT39	APT39 leveraged spearphishing emails with malicious attachments to initially compromise victims. ^{[41][42][43]}
G0096	APT41	APT41 sent spearphishing emails with attachments such as compiled HTML (.chm) files to initially compromise their victims. ^[44]
S0373	Astaroth	Astaroth has been delivered via malicious e-mail attachments. ^[45]
S1087	AsyncRAT	AsyncRAT has been delivered via malicious email attachments. ^[11]
S0642	BADFLICK	BADFLICK has been distributed via spearphishing campaigns containing malicious Microsoft Word documents. ^[46]
S0234	Bandook	Bandook is delivered via a malicious Word document inside a zip file. ^[47]

ID	Name	Description
S0268	Bisonal	Bisonal has been delivered as malicious email attachments. ^[48]
G1002	BITTER	BITTER has sent spearphishing emails with a malicious RTF document or Excel spreadsheet. ^{[49][50]}
G0098	BlackTech	BlackTech has used spearphishing e-mails with malicious password-protected archived files (ZIP or RAR) to deliver malware. ^{[51][52]}
S0520	BLINDINGCAN	BLINDINGCAN has been delivered by phishing emails containing malicious Microsoft Office documents. ^[53]
G0060	BRONZE BUTLER	BRONZE BUTLER used spearphishing emails with malicious Microsoft Word attachments to infect victims. ^{[54][55]}
S1039	Bumblebee	Bumblebee has gained execution through luring users into opening malicious attachments. ^{[56][57][58][59]}
C0011	C0011	During C0011 , Transparent Tribe sent malicious attachments via email to student targets in India. ^[60]
C0015	C0015	For C0015 , security researchers assessed the threat actors likely used a phishing campaign to distribute a weaponized attachment to victims. ^[61]
S0631	Chaes	Chaes has been delivered by sending victims a phishing email containing a malicious .docx file. ^[62]
S0660	Clambling	Clambling has been delivered to victim's machines through malicious e-mail attachments. ^[63]
G0080	Cobalt Group	Cobalt Group has sent spearphishing emails with various attachment types to corporate and personal email accounts of victim organizations. Attachment types have included .rtf, .doc, .xls, archives containing LNK files, and password protected archives containing .exe and .scr executables. ^{[64][65][66][67][68][69][70][71]}
G0142	Confucius	Confucius has crafted and sent victims malicious attachments to gain initial access. ^[72]
G1012	CURIUM	CURIUM has used phishing with malicious attachments for initial access to victim environments. ^[73]
S1014	DanBot	DanBot has been distributed within a malicious Excel attachment via spearphishing emails. ^[74]
S1111	DarkGate	DarkGate can be distributed through emails with malicious attachments from a spoofed email address. ^[75]
G0012	Darkhotel	Darkhotel has sent spearphishing emails with malicious RAR and .LNK attachments. ^{[76][77]}
G0079	DarkHydrus	DarkHydrus has sent spearphishing emails with password-protected RAR archives containing malicious Excel Web Query files (.iqy). The group has also sent spearphishing emails that contained malicious Microsoft Office documents that use the "attachedTemplate" technique to load a template from a remote server. ^{[1][78][79]}
S1066	DarkTortilla	DarkTortilla has been distributed via spearphishing emails containing archive attachments, with file types such as .iso, .zip, .img, .dmg, and .tar, as well as through malicious documents. ^[80]
S0673	DarkWatchman	DarkWatchman has been delivered via spearphishing emails that contain a malicious zip file. ^[81]
G0035	Dragonfly	Dragonfly has sent emails with malicious attachments to gain initial access. ^[82]
G0066	Elderwood	Elderwood has delivered zero-day exploits and malware to victims via targeted emails containing malicious attachments. ^{[83][84]}
S0367	Emotet	Emotet has been delivered by phishing emails containing attachments. ^{[85][86][87][88][89][90][91][92][93]}
S0634	EnvyScout	EnvyScout has been distributed via spearphishing as an email attachment. ^[94]

ID	Name	Description
G1011	EXOTIC LILY	EXOTIC LILY conducted an e-mail thread-hijacking campaign with malicious ISO attachments. ^{[95][56]}
G0137	Ferocious Kitten	Ferocious Kitten has conducted spearphishing campaigns containing malicious documents to lure victims to open the attachments. ^[96]
G0085	FIN4	FIN4 has used spearphishing emails containing attachments (which are often stolen, legitimate documents sent from compromised accounts) with embedded malicious macros. ^{[97][98]}
G0037	FIN6	FIN6 has targeted victims with e-mails containing malicious attachments. ^[99]
G0046	FIN7	FIN7 sent spearphishing emails with either malicious Microsoft Documents or RTF files attached. ^{[100][101][102][103][104]}
G0061	FIN8	FIN8 has distributed targeted emails containing Word documents with embedded malicious macros. ^{[105][106][107]}
S0696	Flagpro	Flagpro has been distributed via spearphishing as an email attachment. ^[52]
C0001	Frankenstein	During Frankenstein , the threat actors likely used spearphishing emails to send malicious Microsoft Word documents. ^[108]
G0084	Gallmaker	Gallmaker sent emails with malicious Microsoft Office documents attached. ^[109]
G0047	Gamaredon Group	Gamaredon Group has delivered spearphishing emails with malicious attachments to targets. ^{[110][111][112][113][114][115][116][117][118]} Additionally, Gamaredon Group has distributed malicious LNK files compressed in ZIP archives. ^[119]
G0078	Gorgon Group	Gorgon Group sent emails to victims with malicious Microsoft Office documents attached. ^[120]
S0499	Hancitor	Hancitor has been delivered via phishing emails with malicious attachments. ^[121]
G0126	Higaisa	Higaisa has sent spearphishing emails containing malicious attachments. ^{[122][123]}
S0483	IcedID	IcedID has been delivered via phishing e-mails with malicious attachments. ^{[124][125]}
G0100	Inception	Inception has used weaponized documents attached to spearphishing emails for reconnaissance and initial compromise. ^{[126][127][128][129]}
G0136	IndigoZebra	IndigoZebra sent spearphishing emails containing malicious password-protected RAR attachments. ^{[130][131]}
S0528	Javali	Javali has been delivered as malicious e-mail attachments. ^[45]
S0648	JSS Loader	JSS Loader has been delivered by phishing emails containing malicious Microsoft Excel attachments. ^[103]
S9044	Kali365	Kali365 has delivered phishing emails with malicious PDF, Word, Excel, and PowerPoint attachments that direct victims to actor-controlled landing pages. ^[132]
S0585	Kerrdown	Kerrdown has been distributed through malicious e-mail attachments. ^[35]
G0094	Kimsuky	Kimsuky has used emails containing Word, Excel and/or HWP (Hangul Word Processor) documents in their spearphishing campaigns. ^{[133][134][135][136][137][8][138][139]} Kimsuky has also distributed emails with attached compressed zip files that contained malicious .LNK files masquerading as legitimate files. ^[140] Kimsuky has delivered tailored PDF documents that contain malicious links. ^[141]
S0669	KOCTOPUS	KOCTOPUS has been distributed via spearphishing emails with malicious attachments. ^[142]
S0356	KONNI	KONNI has been delivered via spearphishing campaigns through a malicious Word document. ^[143]

ID	Name	Description
S1075	KOPILUWAK	KOPILUWAK has been delivered to victims as a malicious email attachment. ^[144]
S9035	LAMEHUG	LAMEHUG has been distributed through spearphishing emails with various AI-themed malicious attachments. ^[145]
S1160	Latrodectus	Latrodectus has been distributed through reply-chain phishing emails with malicious attachments. ^[146]
G0032	Lazarus Group	Lazarus Group has targeted victims with spearphishing emails containing malicious Microsoft Word documents. ^{[147][148][149][150]}
G0140	LazyScripter	LazyScripter has used spam emails weaponized with archive or document files as its initial infection vector. ^[142]
G0065	Leviathan	Leviathan has sent spearphishing emails with malicious attachments, including .rtf, .doc, and .xls files. ^{[151][152]}
S9020	LODEINFO	LODEINFO has been distributed to targeted victims via malicious email attachments. ^{[153][154][155]}
S0447	Lokibot	Lokibot is delivered via a malicious XLS attachment contained within a spearphishing email. ^[156]
S1213	Lumma Stealer	Lumma Stealer has been delivered through phishing emails with malicious attachments. ^[157]
G0095	Machete	Machete has delivered spearphishing emails that contain a zipped file with malicious contents. ^{[158][159][160]}
G1026	Malteiro	Malteiro has sent spearphishing emails containing malicious .zip files. ^[161]
G0045	menuPass	menuPass has sent malicious Office documents via email as part of spearphishing campaigns as well as executables disguised as documents. ^{[162][163][164][165]}
S0455	Metamorfo	Metamorfo has been delivered to victims via emails with malicious HTML attachments. ^{[166][167]}
G1054	MirrorFace	MirrorFace has sent spearphishing emails with malicious attachments to deliver malware payloads. ^{[153][154][155]}
G0103	Mofang	Mofang delivered spearphishing emails with malicious documents, PDFs, or Excel files attached. ^[168]
G0021	Molerats	Molerats has sent phishing emails with malicious Microsoft Word and PDF attachments. ^{[169][170][171]}
G1036	Moonstone Sleet	Moonstone Sleet delivered various payloads to victims as spearphishing attachments. ^[172]
G0069	MuddyWater	MuddyWater has compromised third parties and used compromised accounts to send spearphishing emails with targeted attachments to recipients. ^{[173][174][175][176][177][178][179][180][181][182]} MuddyWater has also sent spearphishing emails with the attachment Cybersecurity.doc, which served as the primarily payload for the next stage. ^[183]
G0129	Mustang Panda	Mustang Panda has used spearphishing attachments to deliver initial access payloads. ^{[184][185][186][187][188][189][190][191][192]} Mustang Panda has also delivered archive files such as RAR and ZIP files containing legitimate EXEs and malicious DLLs. ^{[186][187][188]}
G0019	Naikon	Naikon has used malicious e-mail attachments to deliver malware. ^[193]
S0198	NETWIRE	NETWIRE has been spread via e-mail campaigns utilizing malicious attachments. ^{[194][195]}
G0133	Nomadic Octopus	Nomadic Octopus has targeted victims with spearphishing emails containing malicious attachments. ^{[196][197]}

ID	Name	Description
S0346	OceanSalt	OceanSalt has been delivered via spearphishing emails with Microsoft Office attachments. ^[198]
S0340	Octopus	Octopus has been delivered via spearsphishing emails. ^[197]
G0049	OilRig	OilRig has sent spearphising emails with malicious attachments to potential victims using compromised and/or spoofed email accounts. ^{[199][200][201][202]}
C0060	Operation AkaiRyū	During Operation AkaiRyū , MirrorFace distributed crafted spearphishing emails containing malicious attachments. ^{[203][204]}
C0022	Operation Dream Job	During Operation Dream Job , Lazarus Group sent emails with malicious attachments to gain unauthorized access to targets' computers. ^{[205][206]}
C0016	Operation Dust Storm	During Operation Dust Storm , the threat actors sent spearphishing emails that contained a malicious Microsoft Word document. ^[207]
C0005	Operation Spalax	During Operation Spalax , the threat actors sent phishing emails that included a PDF document that in some cases led to the download and execution of malware. ^[208]
S1017	OutSteel	OutSteel has been distributed as a malicious attachment within a spearphishing email. ^[209]
G0040	Patchwork	Patchwork has used spearphishing with an attachment to deliver files with exploits to initial victims. ^{[210][211][212][213]}
G0068	PLATINUM	PLATINUM has sent spearphishing emails with attachments to victims as its primary initial access vector. ^[214]
S0428	PoetRAT	PoetRAT was distributed via malicious Word documents. ^[215]
S0453	Pony	Pony has been delivered via spearphishing attachments. ^[216]
S0650	QakBot	QakBot has spread through emails with malicious attachments. ^{[217][218][219][220][221][222][223][224][225]}
S1242	Qilin	Qilin has been delivered to victims through malicious email attachments. ^[226]
S0458	Ramsay	Ramsay has been distributed through spearphishing emails with malicious attachments. ^[227]
G0075	Rancor	Rancor has attached a malicious document to an email to gain initial access. ^[228]
G1039	RedCurl	RedCurl has used phishing emails with malicious files to gain initial access. ^{[229][230]}
C0047	RedDelta Modified PlugX Infection Chain Operations	Mustang Panda leveraged malicious attachments in spearphishing emails for initial access to victim environments in RedDelta Modified PlugX Infection Chain Operations . ^[231]
S0332	Remcos	Remcos has been spread through emails containing malicious documents. ^[232]
S0496	REvil	REvil has been distributed via malicious e-mail attachments including MS Word Documents. ^{[233][234][235][236][237]}
S0433	Rifdoor	Rifdoor has been distributed in e-mails with malicious Excel or Word documents. ^[238]
S0240	ROKRAT	ROKRAT has been delivered via spearphishing emails that contain a malicious Hangul Office or Microsoft Word document. ^[239]
S0148	RTM	RTM has been delivered via spearphishing attachments disguised as PDF documents. ^[240]
G0048	RTM	RTM has used spearphishing attachments to distribute its malware. ^[241]

ID	Name	Description
S9037	RustyWater	RustyWater has sent spearphishing emails with the attachment Cybersecurity.doc, which served as the primary payload for the next stage. ^[183]
G1031	Saint Bear	Saint Bear uses a variety of file formats, such as Microsoft Office documents, ZIP archives, PDF documents, and other items as phishing attachments for initial access. ^[209]
S1018	Saint Bot	Saint Bot has been distributed as malicious attachments within spearphishing emails. ^{[242][209]}
G0034	Sandworm Team	Sandworm Team has delivered malicious Microsoft Office and ZIP file attachments via spearphishing emails. ^{[243][244][245][246][247][248]}
G1008	SideCopy	SideCopy has sent spearphishing emails with malicious hta file attachments. ^[249]
G0121	Sidewinder	Sidewinder has sent e-mails with malicious attachments often crafted for specific targets. ^[250]
G0091	Silence	Silence has sent emails with malicious DOCX, CHM, LNK and ZIP attachments. ^{[251][252][253]}
S1086	Snip3	Snip3 has been delivered to victims through malicious e-mail attachments. ^[254]
S1030	Squirrelwaffle	Squirrelwaffle has been distributed via malicious Microsoft Office documents within spam emails. ^[255]
G1033	Star Blizzard	Star Blizzard has sent emails with malicious .pdf files to spread malware. ^[256]
S1183	StrelaStealer	StrelaStealer has been distributed as a spearphishing attachment. ^[257]
S1064	SVCRReady	SVCRReady has been distributed via spearphishing campaigns containing malicious Mircrosoft Word documents. ^[258]
G1018	TA2541	TA2541 has sent phishing emails with malicious attachments for initial access including MS Word documents. ^{[259][260]}
G0062	TA459	TA459 has targeted victims using spearphishing emails with malicious Microsoft Word attachments. ^[261]
G0092	TA505	TA505 has used spearphishing emails with malicious attachments to initially compromise victims. ^{[262][263][264][265][266][267][268][269][270]}
G0127	TA551	TA551 has sent spearphishing attachments with password protected ZIP files. ^{[271][272][273]}
S0011	Taidoor	Taidoor has been delivered through spearphishing emails. ^[274]
G0089	The White Company	The White Company has sent phishing emails with malicious Microsoft Word attachments to victims. ^[275]
G0027	Threat Group-3390	Threat Group-3390 has used e-mail to deliver malicious attachments to victims. ^[63]
S0665	ThreatNeedle	ThreatNeedle has been distributed via a malicious Word document within a spearphishing email. ^[148]
G0131	Tonto Team	Tonto Team has delivered payloads via spearphishing attachments. ^[276]
G0134	Transparent Tribe	Transparent Tribe has sent spearphishing e-mails with attachments to deliver malicious payloads. ^{[277][278][279][280][281]}
S0266	TrickBot	TrickBot has used an email with an Excel sheet containing a malicious macro to deploy the malware ^[282]

ID	Name	Description
G0081	Tropic Trooper	Tropic Trooper sent spearphishing emails that contained malicious Microsoft Office and fake installer file attachments. ^{[283][284][285][286][287]}
S0476	Valak	Valak has been delivered via spearphishing e-mails with password protected ZIP files. ^[271]
S0670	WarzoneRAT	WarzoneRAT has been distributed as a malicious attachment within an email. ^{[288][72]}
C0037	Water Curupira Pikabot Distribution	Water Curupira Pikabot Distribution attached password-protected ZIP archives to deliver Pikabot installers. ^[289]
G0112	Windshift	Windshift has sent spearphishing emails with attachment to harvest credentials and deliver malware. ^[290]
G1035	Winter Vivern	Winter Vivern leverages malicious attachments delivered via email for initial access activity. ^{[291][292][293]}
G0090	WIRTE	WIRTE has sent emails to intended victims with malicious MS Word and Excel attachments. ^[294]
G0102	Wizard Spider	Wizard Spider has used spearphishing attachments to deliver Microsoft documents containing macros or PDFs containing malicious links to download either Emotet , Bokbot , TrickBot , or Bazar . ^{[295][296][297]}
S1065	Woody RAT	Woody RAT has been delivered via malicious Word documents and archive files. ^[298]
S1207	XLoader	XLoader has been delivered as a phishing attachment, including PDFs with embedded links, Word and Excel files, and various archive files (ZIP, RAR, ACE, and ISOs) containing EXE payloads. ^{[299][300]}
S1013	ZxxZ	ZxxZ has been distributed via spearphishing emails, usually containing a malicious RTF or Excel attachment. ^[49]

Mitigations

ID	Mitigation	Description
M1049	Antivirus/Antimalware	Anti-virus can also automatically quarantine suspicious files.
M1047	Audit	Enable auditing and monitoring for email attachments and file transfers to detect and investigate suspicious activity. Regularly review logs for anomalies related to attachments containing potentially malicious content, as well as any attempts to execute or interact with these files. This practice helps identify spearphishing attempts before they can lead to further compromise.
M1031	Network Intrusion Prevention	Network intrusion prevention systems and systems designed to scan and remove malicious email attachments can be used to block activity.
M1021	Restrict Web-Based Content	Block unknown or unused attachments by default that should not be transmitted over email as a best practice to prevent some vectors, such as .scr, .exe, .pif, .cpl, etc. Some email scanning devices can open and analyze compressed and encrypted formats, such as zip and rar that may be used to conceal malicious attachments.
M1054	Software Configuration	Use anti-spoofing and email authentication mechanisms to filter messages based on validity checks of the sender domain (using SPF) and integrity of messages (using DKIM). Enabling these mechanisms within an organization (through policies such as DMARC) may enable recipients (intra-org and cross domain) to perform similar message filtering and validation. ^{[301][302]}
M1018	User Account Management	Apply user account management principles to limit permissions for accounts interacting with email attachments, ensuring that only necessary accounts have the ability to open or execute files. Restricting account privileges reduces the potential impact of malicious attachments by preventing unauthorized execution or spread of malware within the environment.
M1017	User Training	Users can be trained to identify social engineering techniques and spearphishing emails.

Detection Strategy

ID	Name	Analytic ID	Analytic Description
DET0236	Detection Strategy for Spearphishing Attachment across OS Platforms	AN0655	Detection of spearphishing attachments by correlating suspicious email delivery with subsequent file creation and abnormal process execution (e.g., Office spawning PowerShell or CMD). Behavior chain includes inbound email metadata → attachment stored on disk → process execution → outbound network activity.
		AN0656	Phishing attachments executed on Linux systems are detected by linking email logs to file creation in mail directories and subsequent suspicious process execution. Look for unexpected binaries or scripts spawned from user mail directories and anomalous outbound network activity.
		AN0657	Phishing attachment detection on macOS through correlation of Mail app logs, file creation in user directories, and abnormal process execution (e.g., Preview.app or Mail.app spawning Terminal or scripting binaries). Network traffic after attachment interaction is also monitored.

References

1. [Falcone, R., et al. \(2018, July 27\). New Threat Actor Group DarkHydrus Targets Middle East Government. Retrieved August 2, 2018.](#)
2. [Electricity Information Sharing and Analysis Center; SANS Industrial Control Systems. \(2016, March 18\). Analysis of the Cyber Attack on the Ukranian Power Grid: Defense Use Case. Retrieved March 27, 2018.](#)
3. [FireEye Threat Intelligence. \(2015, December 1\). China-based Cyber Threat Group Uses Dropbox for Malware Communications and Targets Hong Kong Media Outlets. Retrieved December 4, 2015.](#)
4. [Arsene, L. \(2020, April 21\). Oil & Gas Spearphishing Campaigns Drop Agent Tesla Spyware in Advance of Historic OPEC+ Deal. Retrieved May 19, 2020.](#)
5. [Check Point Software Technologies. \(2015\). ROCKET KITTEN: A CAMPAIGN WITH 9 LIVES. Retrieved March 16, 2018.](#)
6. [AhnLab. \(2018, June 23\). Targeted attacks by Andariel Threat Group, a subgroup of the Lazarus. Retrieved September 29, 2021.](#)
7. [Jazi, H. \(2021, April 19\). Lazarus APT conceals malicious code within BMP image to drop its RAT . Retrieved September 29, 2021.](#)
8. [Jazi, H. \(2021, June 1\). Kimsuky APT continues to target South Korean government using AppleSeed backdoor. Retrieved June 10, 2021.](#)
9. [QiAnXin Threat Intelligence Center. \(2019, February 18\). APT-C-36: Continuous Attacks Targeting Colombian Government Institutions and Corporations. Retrieved May 5, 2020.](#)
10. [Global Research & Analysis Team, Kaspersky. \(2024, August 19\). BlindEagle flying high in Latin America. Retrieved April 16, 2026.](#)
11. [Insikt Group. \(2025, August 26\). TAG-144's Persistent Grip on South American Organizations. Retrieved April 16, 2026.](#)
12. [Mandiant. \(n.d.\). APT1 Exposing One of China's Cyber Espionage Units. Retrieved July 18, 2016.](#)
13. [Moran, N., Oppenheim, M., Engle, S., & Wartell, R.. \(2014, September 3\). Darwin's Favorite APT Group \[Blog\]. Retrieved November 12, 2014.](#)
14. [Sancho, D., et al. \(2012, May 22\). IXESHE An APT Campaign. Retrieved June 7, 2019.](#)
15. [Ahl, I. \(2017, June 06\). Privileges and Credentials: Phished at the Request of Counsel. Retrieved May 17, 2018.](#)
16. [Lee, B, et al. \(2018, February 28\). Sofacy Attacks Multiple Government Entities. Retrieved March 15, 2018.](#)
17. [Falcone, R. \(2018, March 15\). Sofacy Uses DealersChoice to Target European Government Agency. Retrieved June 4, 2018.](#)
18. [Lee, B., Falcone, R. \(2018, June 06\). Sofacy Group's Parallel Attacks. Retrieved June 18, 2018.](#)
19. [Mueller, R. \(2018, July 13\). Indictment - United States of America vs. VIKTOR BORISOVICH NETYKSHO, et al. Retrieved November 17, 2024.](#)
20. [Kaspersky Lab's Global Research & Analysis Team. \(2018, February 20\). A Slice of 2017 Sofacy Activity. Retrieved November 27, 2018.](#)
21. [Accenture Security. \(2018, November 29\). SNAKEMACKEREL. Retrieved April 15, 2019.](#)
22. [Hacquebord, F., Remorin, L. \(2020, December 17\). Pawn Storm's Lack of Sophistication as a Strategy. Retrieved January 13, 2021.](#)
152. [CISA. \(2021, July 19\). \(AA21-200A\) Joint Cybersecurity Advisory – Tactics, Techniques, and Procedures of Indicted APT40 Actors Associated with China's MSS Hainan State Security Department. Retrieved August 12, 2021.](#)
153. [Ishimaru, S. \(2022, October 31\). APT10: Tracking down LODEINFO 2022, part I. Retrieved April 17, 2026.](#)
154. [Breitenbacher, D. \(2022, December 14\). Unmasking MirrorFace: Operation LiberalFace targeting Japanese political entities. Retrieved April 17, 2026.](#)
155. [ITOCHU. \(2024, January 24\). The Endless Struggle Against APT10: Insights from LODEINFO v0.6.6 - v0.7.3 Analysis. Retrieved April 17, 2026.](#)
156. [Muhammad, I., Unterbrink, H.. \(2021, January 6\). A Deep Dive into Lokibot Infection Chain. Retrieved August 31, 2021.](#)
157. [Cybereason Security Services Team. \(n.d.\). Your Data Is Under New Lummanagement: The Rise of LummaStealer. Retrieved March 22, 2025.](#)
158. [Kaspersky Global Research and Analysis Team. \(2014, August 20\). El Machete. Retrieved September 13, 2019.](#)
159. [ESET. \(2019, July\). MACHETE JUST GOT SHARPER Venezuelan government institutions under attack. Retrieved September 13, 2019.](#)
160. [kate. \(2020, September 25\). APT-C-43 steals Venezuelan military secrets to provide intelligence support for the reactionaries – HpReact campaign. Retrieved November 20, 2020.](#)
161. [SCILabs. \(2021, December 23\). Cyber Threat Profile Malteiro. Retrieved March 13, 2024.](#)
162. [PwC and BAE Systems. \(2017, April\). Operation Cloud Hopper: Technical Annex. Retrieved April 13, 2017.](#)
163. [FireEye iSIGHT Intelligence. \(2017, April 6\). APT10 \(MenuPass Group\): New Tools, Global Campaign Latest Manifestation of Longstanding Threat. Retrieved June 29, 2017.](#)
164. [Matsuda, A., Muhammad I. \(2018, September 13\). APT10 Targeting Japanese Corporations Using Updated TTPs. Retrieved September 17, 2018.](#)
165. [US District Court Southern District of New York. \(2018, December 17\). United States v. Zhu Hua Indictment. Retrieved December 17, 2020.](#)
166. [Sierra, E., Iglesias, G.. \(2018, April 24\). Metamorfo Campaigns Targeting Brazilian Users. Retrieved July 30, 2020.](#)
167. [ESET Research. \(2019, October 3\). Casbaneiro: peculiarities of this banking Trojan that affects Brazil and Mexico. Retrieved September 23, 2021.](#)
168. [Yonathan Klijnsma. \(2016, May 17\). Mofang: A politically motivated information stealing adversary. Retrieved May 12, 2020.](#)
169. [GReAT. \(2019, April 10\). Gaza Cybergang Group1, operation SneakyPastes. Retrieved May 13, 2020.](#)
170. [Falcone, R., et al. \(2020, March 3\). Molerats Delivers Spark Backdoor to Government and Telecommunications Organizations. Retrieved December 14, 2020.](#)
171. [Cybereason Nocturnus Team. \(2020, December 9\). MOLERATS IN THE CLOUD: New Malware Arsenal Abuses Cloud Platforms in Middle East Espionage Campaign. Retrieved December 22, 2020.](#)
172. [Microsoft Threat Intelligence. \(2024, May 28\). Moonstone Sleet emerges as new North Korean threat actor with new bag of tricks. Retrieved August 26, 2024.](#)

23. [Secureworks CTU. \(2017, March 30\). IRON TWILIGHT Supports Active Measures. Retrieved February 28, 2022.](#)
24. [Simonovich, V. \(2025, July 23\). Cato CTRL™ Threat Research: Analyzing LAMEHUG – First Known LLM-Powered Malware with Links to APT28 \(Fancy Bear\) . Retrieved April 21, 2026.](#)
25. [F-Secure Labs. \(2015, September 17\). The Dukes: 7 years of Russian cyberespionage. Retrieved December 10, 2015.](#)
26. [Microsoft Threat Intelligence Center \(MSTIC\). \(2021, May 27\). New sophisticated email-based attack from NOBELIUM. Retrieved May 28, 2021.](#)
27. [ESET. \(2022, February\). THREAT REPORT T3 2021. Retrieved February 10, 2022.](#)
28. [Secureworks CTU. \(n.d.\). IRON HEMLOCK. Retrieved February 22, 2022.](#)
29. [FireEye Labs. \(2015, April\). APT30 AND THE MECHANICS OF A LONG-RUNNING CYBER ESPIONAGE OPERATION. Retrieved November 17, 2024.](#)
30. [Foltýn, T. \(2018, March 13\). OceanLotus ships new backdoor using old tricks. Retrieved May 22, 2018.](#)
31. [Dahan, A. \(2017, May 24\). OPERATION COBALT KITTY: A LARGE-SCALE APT IN ASIA CARRIED OUT BY THE OCEANLOTUS GROUP. Retrieved November 5, 2018.](#)
32. [Dahan, A. \(2017\). Operation Cobalt Kitty. Retrieved December 27, 2018.](#)
33. [Dumont, R. \(2019, March 20\). Fake or Fake: Keeping up with OceanLotus decoys. Retrieved April 1, 2019.](#)
34. [Henderson, S., et al. \(2020, April 22\). Vietnamese Threat Actors APT32 Targeting Wuhan Government and Chinese Ministry of Emergency Management in Latest Example of COVID-19 Related Espionage. Retrieved April 28, 2020.](#)
35. [Amnesty International. \(2021, February 24\). Vietnamese activists targeted by notorious hacking group. Retrieved March 1, 2021.](#)
36. [Microsoft Threat Protection Intelligence Team. \(2020, June 18\). Inside Microsoft Threat Protection: Mapping attack chains from cloud to endpoint. Retrieved June 22, 2020.](#)
37. [FireEye. \(2018, February 20\). APT37 \(Reaper\): The Overlooked North Korean Actor. Retrieved November 17, 2024.](#)
38. [Mercer, W., Rascagneres, P. \(2018, January 16\). Korea In The Crosshairs. Retrieved May 21, 2018.](#)
39. [GReAT. \(2019, May 13\). ScarCruft continues to evolve, introduces Bluetooth harvester. Retrieved June 4, 2019.](#)
40. [DHS/CISA. \(2020, August 26\). FASTCash 2.0: North Korea's BeagleBoyz Robbing Banks. Retrieved September 29, 2021.](#)
41. [Hawley et al. \(2019, January 29\). APT39: An Iranian Cyber Espionage Group Focused on Personal Information. Retrieved February 19, 2019.](#)
42. [Symantec. \(2018, February 28\). Chafer: Latest Attacks Reveal Heightened Ambitions. Retrieved May 22, 2020.](#)
43. [FBI. \(2020, September 17\). Indicators of Compromise Associated with Rana Intelligence Computing, also known as Advanced Persistent Threat 39, Chafer, Cadelspy, Remexi, and ITG07. Retrieved December 10, 2020.](#)
44. [Fraser, N., et al. \(2019, August 7\). Double Dragon APT41, a dual espionage and cyber crime operation APT41. Retrieved September 23, 2019.](#)
45. [GReAT. \(2020, July 14\). The Tetrade: Brazilian banking malware goes global. Retrieved November 9, 2020.](#)
46. [Accenture iDefense Unit. \(2019, March 5\). Mudcarp's Focus on Submarine Technologies. Retrieved August 24, 2021.](#)
47. [Check Point. \(2020, November 26\). Bandoor: Signed & Delivered. Retrieved May 31, 2021.](#)
173. [Lancaster, T.. \(2017, November 14\). Muddying the Water: Targeted Attacks in the Middle East. Retrieved March 15, 2018.](#)
174. [Singh, S. et al.. \(2018, March 13\). Iranian Threat Group Updates Tactics, Techniques and Procedures in Spear Phishing Campaign. Retrieved April 11, 2018.](#)
175. [Kaspersky Lab's Global Research & Analysis Team. \(2018, October 10\). MuddyWater expands operations. Retrieved November 2, 2018.](#)
176. [ClearSky. \(2019, June\). Iranian APT group 'MuddyWater' Adds Exploits to Their Arsenal. Retrieved May 14, 2020.](#)
177. [Mele, G. et al. \(2021, February 10\). Probable Iranian Cyber Actors, Static Kitten, Conducting Cyberespionage Campaign Targeting UAE and Kuwait Government Agencies. Retrieved March 17, 2021.](#)
178. [Peretz, A. and Theck, E. \(2021, March 5\). Earth Vetala – MuddyWater Continues to Target Organizations in the Middle East. Retrieved March 18, 2021.](#)
179. [FBI, CISA, CNMF, NCSC-UK. \(2022, February 24\). Iranian Government-Sponsored Actors Conduct Cyber Operations Against Global Government and Commercial Networks. Retrieved September 27, 2022.](#)
180. [Miller, J. et al. \(2024, March 21\). Security Brief: TA450 Uses Embedded Links in PDF Attachments in Latest Campaign. Retrieved March 27, 2024.](#)
181. [ESET Research. \(2025, December 2\). MuddyWater: Snakes by the riverbank. Retrieved February 17, 2026.](#)
182. [SOCRadar. \(2026, March 9\). MuddyWater Uses Dindoor Malware Targeting U.S. Networks. Retrieved March 12, 2026.](#)
183. [Awasthi, P. \(2026, January 8\). Reborn in Rust: Muddy Water Evolves Tooling with RustyWater Implant. Retrieved March 19, 2026.](#)
184. [Asheer Malhotra, Jungsoo An, Kendall Mc. \(2022, May 5\). Mustang Panda deploys a new wave of malware targeting Europe. Retrieved August 4, 2025.](#)
185. [CSIRT CTI. \(2024, January 23\). Stately Taurus Targets Myanmar Amidst Concerns over Military Junta's Handling of Rebel Attacks. Retrieved August 4, 2025.](#)
186. [Dex. \(n.d.\). New Mustang Panda's campaigning against Australia. Retrieved August 4, 2025.](#)
187. [Golo Muhr, Joshua Chung. \(2025, June 23\). Hive0154 aka Mustang Panda shifts focus on Tibetan community to deploy Pubload backdoor. Retrieved August 4, 2025.](#)
188. [Golo Muhr, Joshua Chung. \(2025, May 15\). Hive0154 targeting US, Philippines, Pakistan and Taiwan in suspected espionage campaign. Retrieved August 4, 2025.](#)
189. [Huntley, S. \(2022, March 7\). An update on the threat landscape. Retrieved March 16, 2022.](#)
190. [Insikt Group. \(2020, July 28\). CHINESE STATE-SPONSORED GROUP 'REDDELTA' TARGETS THE VATICAN AND CATHOLIC ORGANIZATIONS. Retrieved April 13, 2021.](#)
191. [Lenart Bermejo, Sunny Lu, Ted Lee. \(2024, September 9\). Earth Preta Evolves its Attacks with New Malware and Strategies. Retrieved August 4, 2025.](#)
192. [Proofpoint Threat Research Team. \(2020, November 23\). TA416 Goes to Ground and Returns with a Golang PlugX Malware Loader. Retrieved April 13, 2021.](#)
193. [CheckPoint. \(2020, May 7\). Naikon APT: Cyber Espionage Reloaded. Retrieved May 26, 2020.](#)
194. [Duncan, B. \(2020, April 3\). GuLoader: Malspam Campaign Installing NetWire RAT. Retrieved January 7, 2021.](#)
195. [Proofpoint. \(2020, December 2\). Geofenced NetWire Campaigns. Retrieved January 7, 2021.](#)

48. [Mercer, W., et al. \(2020, March 5\). Bisonal: 10 years of play. Retrieved January 26, 2022.](#)
49. [Raghuprasad, C. \(2022, May 11\). Bitter APT adds Bangladesh to their targets. Retrieved June 1, 2022.](#)
50. [Dela Paz, R. \(2016, October 21\). BITTER: a targeted attack against Pakistan. Retrieved June 1, 2022.](#)
51. [Bermejo, L., et al. \(2017, June 22\). Following the Trail of BlackTech's Cyber Espionage Campaigns. Retrieved May 5, 2020.](#)
52. [Hada, H. \(2021, December 28\). Flagpro The new malware used by BlackTech. Retrieved March 25, 2022.](#)
53. [US-CERT. \(2020, August 19\). MAR-10295134-1.v1 – North Korean Remote Access Trojan: BLINDINGCAN. Retrieved August 19, 2020.](#)
54. [DiMaggio, J. \(2016, April 28\). Tick cyberespionage group zeros in on Japan. Retrieved July 16, 2018.](#)
55. [Chen, J. et al. \(2019, November\). Operation ENDTRADE: TICK's Multi-Stage Backdoors for Attacking Industries and Stealing Classified Data. Retrieved June 9, 2020.](#)
56. [Merriman, K. and Trouerbach, P. \(2022, April 28\). This isn't Optimus Prime's Bumblebee but it's Still Transforming. Retrieved August 22, 2022.](#)
57. [Kamble, V. \(2022, June 28\). Bumblebee: New Loader Rapidly Assuming Central Position in Cyber-crime Ecosystem. Retrieved August 24, 2022.](#)
58. [Cybereason. \(2022, August 17\). Bumblebee Loader – The High Road to Enterprise Domain Control. Retrieved August 29, 2022.](#)
59. [Salem, A. \(2022, April 27\). The chronicles of Bumblebee: The Hook, the Bee, and the Trickbot connection. Retrieved September 2, 2022.](#)
60. [N. Baisini. \(2022, July 13\). Transparent Tribe begins targeting education sector in latest campaign. Retrieved September 22, 2022.](#)
61. [DFIR Report. \(2021, November 29\). CONTInuing the Bazar Ransomware Story. Retrieved September 29, 2022.](#)
62. [Salem, E. \(2020, November 17\). CHAES: Novel Malware Targeting Latin American E-Commerce. Retrieved June 30, 2021.](#)
63. [Lunghi, D. et al. \(2020, February\). Uncovering DRBControl. Retrieved November 12, 2021.](#)
64. [Svajcer, V. \(2018, July 31\). Multiple Cobalt Personality Disorder. Retrieved September 5, 2018.](#)
65. [Positive Technologies. \(2017, August 16\). Cobalt Strikes Back: An Evolving Multinational Threat to Finance. Retrieved September 5, 2018.](#)
66. [Positive Technologies. \(2016, December 16\). Cobalt Snatch. Retrieved October 9, 2018.](#)
67. [Matveeva, V. \(2017, August 15\). Secrets of Cobalt. Retrieved October 10, 2018.](#)
68. [Mesa, M, et al. \(2017, June 1\). Microsoft Word Intruder Integrates CVE-2017-0199, Utilized by Cobalt Group to Target Financial Institutions. Retrieved October 10, 2018.](#)
69. [Klijnsma, Y.. \(2017, November 28\). Gaffe Reveals Full List of Targets in Spear Phishing Attack Using Cobalt Strike Against Financial Institutions. Retrieved October 10, 2018.](#)
70. [Unit 42. \(2018, October 25\). New Techniques to Uncover and Attribute Financial actors Commodity Builders and Infrastructure Revealed. Retrieved December 11, 2018.](#)
71. [Giagone, R., Bermejo, L., and Yarochkin, F. \(2017, November 20\). Cobalt Strikes Again: Spam Runs Use Macros and CVE-2017-8759 Exploit Against Russian Banks. Retrieved March 7, 2019.](#)
196. [Paganini, P. \(2018, October 16\). Russia-linked APT group DustSquad targets diplomatic entities in Central Asia. Retrieved August 24, 2021.](#)
197. [Cherepanov, A. \(2018, October 4\). Nomadic Octopus Cyber espionage in Central Asia. Retrieved October 13, 2021.](#)
198. [Sherstobitoff, R., Malhotra, A. \(2018, October 18\). 'Operation Oceansalt' Attacks South Korea, U.S., and Canada With Source Code From Chinese Hacker Group. Retrieved November 30, 2018.](#)
199. [Lee, B., Falcone, R. \(2018, February 23\). Oops! OilRig Uses ThreeDollars to Deliver New Trojan. Retrieved July 16, 2018.](#)
200. [Lee, B., Falcone, R. \(2018, July 25\). OilRig Targets Technology Service Provider and Government Agency with QUADAGENT. Retrieved August 9, 2018.](#)
201. [Meyers, A. \(2018, November 27\). Meet CrowdStrike's Adversary of the Month for November: HELIX KITTEN. Retrieved December 18, 2018.](#)
202. [ClearSky Cybersecurity. \(2017, January 5\). Iranian Threat Agent OilRig Delivers Digitally Signed Malware, Impersonates University of Oxford. Retrieved May 3, 2017.](#)
203. [Dominik Breitenbacher. \(2025, March 18\). Operation AkaiRyū: MirrorFace invites Europe to Expo 2025 and revives ANEL backdoor. Retrieved May 22, 2025.](#)
204. [Hiroaki, H. \(2024, November 26\). Guess Who's Back - The Return of ANEL in the Recent Earth Kasha Spear-phishing Campaign in 2024. Retrieved April 17, 2026.](#)
205. [ClearSky Research Team. \(2020, August 13\). Operation 'Dream Job' Widespread North Korean Espionage Campaign. Retrieved December 20, 2021.](#)
206. [Cashman, M. \(2020, July 29\). Operation North Star Campaign. Retrieved December 20, 2021.](#)
207. [Gross, J. \(2016, February 23\). Operation Dust Storm. Retrieved December 22, 2021.](#)
208. [M. Porolli. \(2021, January 21\). Operation Spalax: Targeted malware attacks in Colombia. Retrieved September 16, 2022.](#)
209. [Unit 42. \(2022, February 25\). Spear Phishing Attacks Target Organizations in Ukraine, Payloads Include the Document Stealer OutSteel and the Downloader SaintBot. Retrieved June 9, 2022.](#)
210. [Cymmetria. \(2016\). Unveiling Patchwork - The Copy-Paste APT. Retrieved November 17, 2024.](#)
211. [Kaspersky Lab's Global Research & Analysis Team. \(2016, July 8\). The Dropping Elephant – aggressive cyber-espionage in the Asian region. Retrieved August 3, 2016.](#)
212. [Lunghi, D., et al. \(2017, December\). Untangling the Patchwork Cyberespionage Group. Retrieved July 10, 2018.](#)
213. [Meltzer, M, et al. \(2018, June 07\). Patchwork APT Group Targets US Think Tanks. Retrieved July 16, 2018.](#)
214. [Windows Defender Advanced Threat Hunting Team. \(2016, April 29\). PLATINUM: Targeted attacks in South and Southeast Asia. Retrieved February 15, 2018.](#)
215. [Mercer, W, et al. \(2020, April 16\). PoetRAT: Python RAT uses COVID-19 lures to target Azerbaijan public and private sectors. Retrieved April 27, 2020.](#)
216. [hasherezade. \(2016, April 11\). No money, but Pony! From a mail to a trojan horse. Retrieved May 21, 2020.](#)
217. [Mendoza, E. et al. \(2020, May 25\). Qakbot Resurges, Spreads through VBS Files. Retrieved September 27, 2021.](#)
218. [Sette, N. et al. \(2020, June 4\). Qakbot Malware Now Exfiltrating Emails for Sophisticated Thread Hijacking Attacks. Retrieved September 27, 2021.](#)
219. [Trend Micro. \(2020, December 17\). QAKBOT: A decade-old malware still with new tricks. Retrieved November 17, 2024.](#)

72. [Uptycs Threat Research Team. \(2021, January 12\). Confucius APT deploys Warzone RAT. Retrieved December 17, 2021.](#)
73. [PwC Threat Intelligence. \(2023, October 25\). Yellow Liderc ships its scripts and delivers IMAPLoader malware. Retrieved August 14, 2024.](#)
74. [SecureWorks 2019, August 27 LYCEUM Takes Center Stage in Middle East Campaign Retrieved. 2019/11/19](#)
75. [Adi Zeligson & Rotem Kerner. \(2018, November 13\). Enter The DarkGate - New Cryptocurrency Mining and Ransomware Campaign. Retrieved February 9, 2024.](#)
76. [Kaspersky Lab's Global Research & Analysis Team. \(2015, August 10\). Darkhotel's attacks in 2015. Retrieved November 2, 2018.](#)
77. [Microsoft. \(2016, July 14\). Reverse engineering DUBNIUM – Stage 2 payload analysis . Retrieved March 31, 2021.](#)
78. [Falcone, R. \(2018, August 07\). DarkHydrus Uses Phishery to Harvest Credentials in the Middle East. Retrieved August 10, 2018.](#)
79. [Unit 42. \(2017, December 15\). Unit 42 Playbook Viewer. Retrieved December 20, 2017.](#)
80. [Secureworks Counter Threat Unit Research Team. \(2022, August 17\). DarkTortilla Malware Analysis. Retrieved November 3, 2022.](#)
81. [Smith, S., Stafford, M. \(2021, December 14\). DarkWatchman: A new evolution in fileless techniques. Retrieved January 10, 2022.](#)
82. [Slowik, J. \(2021, October\). THE BAFFLING BERSERK BEAR: A DECADE'S ACTIVITY TARGETING CRITICAL INFRASTRUCTURE. Retrieved December 6, 2021.](#)
83. [O'Gorman, G., and McDonald, G.. \(2012, September 6\). The Elderwood Project. Retrieved November 17, 2024.](#)
84. [Clayton, M.. \(2012, September 14\). Stealing US business secrets: Experts ID two huge cyber 'gangs' in China. Retrieved February 15, 2018.](#)
85. [CIS. \(2017, April 28\). Emotet Changes TTPs and Arrives in United States. Retrieved January 17, 2019.](#)
86. [Smith, A.. \(2017, December 22\). Protect your network from Emotet Trojan with Malwarebytes Endpoint Security. Retrieved January 17, 2019.](#)
87. [Symantec. \(2018, July 18\). The Evolution of Emotet: From Banking Trojan to Threat Distributor. Retrieved March 25, 2019.](#)
88. [US-CERT. \(2018, July 20\). Alert \(TA18-201A\) Emotet Malware. Retrieved March 25, 2019.](#)
89. [Brumaghin, E.. \(2019, January 15\). Emotet re-emerges after the holidays. Retrieved March 25, 2019.](#)
90. [Trend Micro. \(2019, January 16\). Exploring Emotet's Activities . Retrieved March 25, 2019.](#)
91. [Özarslan, S. \(2018, December 21\). The Christmas Card you never wanted - A new wave of Emotet is back to wreak havoc. Retrieved March 25, 2019.](#)
92. [Lee, S.. \(2019, April 24\). Emotet Using WMI to Launch PowerShell Encoded Code. Retrieved May 24, 2019.](#)
93. [Kessem, L., et al. \(2017, November 13\). New Banking Trojan IcedID Discovered by IBM X-Force Research. Retrieved July 14, 2020.](#)
94. [MSTIC. \(2021, May 28\). Breaking down NOBELIUM's latest early-stage toolset. Retrieved August 4, 2021.](#)
95. [Stolyarov, V. \(2022, March 17\). Exposing initial access broker with ties to Conti. Retrieved August 18, 2022.](#)
96. [GReAT. \(2021, June 16\). Ferocious Kitten: 6 Years of Covert Surveillance in Iran. Retrieved September 22, 2021.](#)
220. [Cyberint. \(2021, May 25\). Qakbot Banking Trojan. Retrieved September 27, 2021.](#)
221. [Morrow, D. \(2021, April 15\). The rise of QakBot. Retrieved September 27, 2021.](#)
222. [Kuzmenko, A. et al. \(2021, September 2\). QakBot technical analysis. Retrieved September 27, 2021.](#)
223. [Group IB. \(2020, September\). LOCK LIKE A PRO. Retrieved November 17, 2024.](#)
224. [Vilkomir-Preisman, S. \(2022, August 18\). Beating Black Basta Ransomware. Retrieved March 8, 2023.](#)
225. [Microsoft. \(2022, May 9\). Ransomware as a service: Understanding the cybercrime gig economy and how to protect yourself. Retrieved March 10, 2023.](#)
226. [SentinelOne. \(2022, November 30\). Agenda \(Qilin\). Retrieved September 26, 2025.](#)
227. [Antiy CERT. \(2020, April 20\). Analysis of Ramsay components of Darkhotel's infiltration and isolation network. Retrieved March 24, 2021.](#)
228. [Ash, B., et al. \(2018, June 26\). RANCOR: Targeted Attacks in South East Asia Using PLAINTIEE and DDKONG Malware Families. Retrieved July 2, 2018.](#)
229. [Group-IB. \(2020, August\). RedCurl: The Pentest You Didn't Know About. Retrieved August 9, 2024.](#)
230. [Tancio et al. \(2024, March 6\). Unveiling Earth Kapre aka RedCurl's Cyberespionage Tactics With Trend Micro MDR, Threat Intelligence. Retrieved August 9, 2024.](#)
231. [Insikt Group. \(2025, January 9\). Chinese State-Sponsored RedDelta Targeted Taiwan, Mongolia, and Southeast Asia with Adapted PlugX Infection Chain. Retrieved January 14, 2025.](#)
232. [Zhang, X. \(2024, November 8\). New Campaign Uses Remcos RAT to Exploit Victims. Retrieved April 16, 2026.](#)
233. [Han, Karsten. \(2019, June 4\). Strange Bits: Sodinokibi Spam, CinaRAT, and Fake G DATA. Retrieved August 4, 2020.](#)
234. [Cylance. \(2019, July 3\). hreat Spotlight: Sodinokibi Ransomware. Retrieved August 4, 2020.](#)
235. [Counter Threat Unit Research Team. \(2019, September 24\). REvil/Sodinokibi Ransomware. Retrieved August 4, 2020.](#)
236. [McAfee. \(2019, October 2\). McAfee ATR Analyzes Sodinokibi aka REvil Ransomware-as-a-Service – What The Code Tells Us. Retrieved August 4, 2020.](#)
237. [Ozarslan, S. \(2020, January 15\). A Brief History of Sodinokibi. Retrieved August 5, 2020.](#)
238. [Knight, S.. \(2020, April 16\). VMware Carbon Black TAU Threat Analysis: The Evolution of Lazarus. Retrieved May 1, 2020.](#)
239. [Jazi, Hossein. \(2021, January 6\). Retrohunting APT37: North Korean APT used VBA self decode technique to inject RokRat. Retrieved March 22, 2022.](#)
240. [Duncan, B., Harbison, M. \(2019, January 23\). Russian Language Malspam Pushing Redaman Banking Malware. Retrieved June 16, 2020.](#)
241. [Skulkin, O. \(2019, August 5\). Following the RTM Forensic examination of a computer infected with a banking trojan. Retrieved May 11, 2020.](#)
242. [Hasherezade. \(2021, April 6\). A deep dive into Saint Bot, a new downloader. Retrieved June 9, 2022.](#)
243. [Ward, S.. \(2014, October 14\). iSIGHT discovers zero-day vulnerability CVE-2014-4114 used in Russian cyber-espionage campaign. Retrieved November 17, 2024.](#)
244. [US-CERT. \(2016, February 25\). ICS Alert \(IR-ALERT-H-16-056-01\) Cyber-Attack Against Ukrainian Critical Infrastructure. Retrieved June 10, 2020.](#)
245. [Cherepanov, A.. \(2016, December 13\). The rise of TeleBots: Analyzing disruptive KillDisk attacks. Retrieved June 10, 2020.](#)

97. [Vengerik, B. et al.. \(2014, December 5\). Hacking the Street? FIN4 Likely Playing the Market. Retrieved December 17, 2018.](#)
98. [Vengerik, B. & Dennesen, K.. \(2014, December 5\). Hacking the Street? FIN4 Likely Playing the Market. Retrieved January 15, 2019.](#)
99. [Visa Public. \(2019, February\). FIN6 Cybercrime Group Expands Threat to eCommerce Merchants. Retrieved September 16, 2019.](#)
100. [Carr, N., et al. \(2017, April 24\). FIN7 Evolution and the Phishing LNK. Retrieved April 24, 2017.](#)
101. [Department of Justice. \(2018, August 01\). HOW FIN7 ATTACKED AND STOLE DATA. Retrieved August 24, 2018.](#)
102. [Platt, J. and Reeves, J.. \(2019, March\). FIN7 Revisited: Inside Astra Panel and SQLRat Malware. Retrieved June 18, 2019.](#)
103. [eSentire. \(2021, July 21\). Notorious Cybercrime Gang, FIN7, Lands Malware in Law Firm Using Fake Legal Complaint Against Jack Daniels' Owner, Brown-Forman Inc.. Retrieved September 20, 2021.](#)
104. [Loui, E. and Reynolds, J. \(2021, August 30\). CARBON SPIDER Embraces Big Game Hunting, Part 1. Retrieved September 20, 2021.](#)
105. [Bohannon, D. & Carr N. \(2017, June 30\). Obfuscation in the Wild: Targeted Attackers Lead the Way in Evasion Techniques. Retrieved February 12, 2018.](#)
106. [Kizhakkinan, D., et al. \(2016, May 11\). Threat Actor Leverages Windows Zero-day Exploit in Payment Card Data Attacks. Retrieved February 12, 2018.](#)
107. [Elovitz, S. & Ahl, I. \(2016, August 18\). Know Your Enemy: New Financially-Motivated & Spear-Phishing Group. Retrieved February 26, 2018.](#)
108. [Adamitis, D. et al. \(2019, June 4\). It's alive: Threat actors cobble together open-source pieces into monstrous Frankenstein campaign. Retrieved May 11, 2020.](#)
109. [Symantec Security Response. \(2018, October 10\). Gallmaker: New Attack Group Eschews Malware to Live off the Land. Retrieved November 27, 2018.](#)
110. [Kakara, H., Maruyama, E. \(2020, April 17\). Gamaredon APT Group Use Covid-19 Lure in Campaigns. Retrieved May 19, 2020.](#)
111. [Boutin, J. \(2020, June 11\). Gamaredon group grows its game. Retrieved June 16, 2020.](#)
112. [CERT-EE. \(2021, January 27\). Gamaredon Infection: From Dropper to Entry. Retrieved February 17, 2022.](#)
113. [Microsoft Threat Intelligence Center. \(2022, February 4\). ACTINIUM targets Ukrainian organizations. Retrieved February 18, 2022.](#)
114. [Unit 42. \(2022, February 3\). Russia's Gamaredon aka Primitive Bear APT Group Actively Targeting Ukraine. Retrieved February 21, 2022.](#)
115. [Secureworks CTU. \(n.d.\). IRON TILDEN. Retrieved February 24, 2022.](#)
116. [Unit 42. \(2022, December 20\). Russia's Trident Ursa \(aka Gamaredon APT\) Cyber Conflict Operations Unwavering Since Invasion of Ukraine. Retrieved September 12, 2024.](#)
117. [Rusnák, Z. \(2024, September 26\). Cyberespionage the Gamaredon way: Analysis of toolset used to spy on Ukraine in 2022 and 2023. Retrieved October 30, 2024.](#)
118. [Silent Push. \(2023, September 7\). From Russia with a 71: Uncovering Gamaredon's fast flux infrastructure. New Apex domains and ASN/IP diversity patterns discovered. Retrieved July 28, 2025.](#)
246. [Scott W. Brady. \(2020, October 15\). United States vs. Yuriy Sergeyevich Andrienko et al.. Retrieved November 25, 2020.](#)
247. [Morgan, K. \(2023, October 18\). Government-backed actors exploiting WinRAR vulnerability. Retrieved July 19, 2024.](#)
248. [Roncone, G. et al. \(n.d.\). APT44: Unearthing Sandworm. Retrieved July 11, 2024.](#)
249. [Threat Intelligence Team. \(2021, December 2\). SideCopy APT: Connecting lures victims, payloads to infrastructure. Retrieved June 13, 2022.](#)
250. [Hegel, T. \(2021, January 13\). A Global Perspective of the SideWinder APT. Retrieved January 27, 2021.](#)
251. [Skulkin, O.. \(2019, January 20\). Silence: Dissecting Malicious CHM Files and Performing Forensic Analysis. Retrieved November 17, 2024.](#)
252. [GReAT. \(2017, November 1\). Silence – a new Trojan attacking financial organizations. Retrieved May 24, 2019.](#)
253. [Group-IB. \(2018, September\). Silence: Moving Into the Darkside. Retrieved May 5, 2020.](#)
254. [Jornet, A. \(2021, December 23\). Snip3, an investigation into malware. Retrieved September 19, 2023.](#)
255. [Palazolo, G. \(2021, October 7\). SquirrelWaffle: New Malware Loader Delivering Cobalt Strike and QakBot. Retrieved August 9, 2022.](#)
256. [Shields, W. \(2024, January 18\). Russian threat group COLDRIVER expands its targeting of Western officials to include the use of malware. Retrieved June 13, 2024.](#)
257. [DCSO CyTec Blog. \(2022, November 8\). #ShortAndMalicious: StrelaStealer aims for mail credentials. Retrieved December 31, 2024.](#)
258. [Schlapfer, Patrick. \(2022, June 6\). A New Loader Gets Ready. Retrieved December 13, 2022.](#)
259. [Larson, S. and Wise, J. \(2022, February 15\). Charting TA2541's Flight. Retrieved September 12, 2023.](#)
260. [Ventura, V. \(2021, September 16\). Operation Layover: How we tracked an attack on the aviation industry to five years of compromise. Retrieved September 15, 2023.](#)
261. [Axel F. \(2017, April 27\). APT Targets Financial Analysts with CVE-2017-0199. Retrieved February 15, 2018.](#)
262. [Proofpoint Staff. \(2017, September 27\). Threat Actor Profile: TA505, From Dridex to Globelmposter. Retrieved May 28, 2019.](#)
263. [Proofpoint Staff. \(2018, June 8\). TA505 shifts with the times. Retrieved May 28, 2019.](#)
264. [Schwarz, D. and Proofpoint Staff. \(2019, January 9\). ServHelper and FlawedGrace - New malware introduced by TA505. Retrieved May 28, 2019.](#)
265. [Salem, E. \(2019, April 25\). Threat Actor TA505 Targets Financial Enterprises Using LOLBins and a New Backdoor Malware. Retrieved May 28, 2019.](#)
266. [Proofpoint Staff. \(2018, July 19\). TA505 Abusing SettingContent-ms within PDF files to Distribute FlawedAmmy RAT. Retrieved April 19, 2019.](#)
267. [Proofpoint Staff. \(2018, March 7\). Leaked Ammy Admin Source Code Turned into Malware. Retrieved May 28, 2019.](#)
268. [Hiroaki, H. and Lu, L. \(2019, June 12\). Shifting Tactics: Breaking Down TA505 Group's Use of HTML, RATs and Other Techniques in Latest Campaigns. Retrieved May 29, 2020.](#)
269. [Schwarz, D. et al. \(2019, October 16\). TA505 Distributes New SDBbot Remote Access Trojan with Get2 Downloader. Retrieved May 29, 2020.](#)
270. [Frydrych, M. \(2020, April 14\). TA505 Continues to Infect Networks With SDBbot RAT. Retrieved May 29, 2020.](#)

119. [Venere, G. \(2025, March 28\). Gamaredon campaign abuses LNK files to distribute Remcos backdoor. Retrieved July 23, 2025.](#)
120. [Falcone, R., et al. \(2018, August 02\). The Gorgon Group: Slithering Between Nation State and Cybercrime. Retrieved August 7, 2018.](#)
121. [Anubhav, A., Jallepalli, D. \(2016, September 23\). Hancitor \(AKA Chanitor\) observed using multiple attack approaches. Retrieved August 13, 2020.](#)
122. [Malwarebytes Threat Intelligence Team. \(2020, June 4\). New LNK attack tied to Higaisa APT discovered. Retrieved March 2, 2021.](#)
123. [Singh, S. Singh, A. \(2020, June 11\). The Return on the Higaisa APT. Retrieved March 2, 2021.](#)
124. [Kimayong, P. \(2020, June 18\). COVID-19 and FMLA Campaigns used to install new IcedID banking malware. Retrieved July 14, 2020.](#)
125. [DFIR. \(2021, March 29\). Sodinokibi \(aka REvil\) Ransomware. Retrieved July 22, 2024.](#)
126. [GReAT. \(2014, December 10\). Cloud Atlas: RedOctober APT is back in style. Retrieved May 8, 2020.](#)
127. [Symantec. \(2018, March 14\). Inception Framework: Alive and Well, and Hiding Behind Proxies. Retrieved May 8, 2020.](#)
128. [Lancaster, T. \(2018, November 5\). Inception Attackers Target Europe with Year-old Office Vulnerability. Retrieved May 8, 2020.](#)
129. [GReAT. \(2019, August 12\). Recent Cloud Atlas activity. Retrieved May 8, 2020.](#)
130. [Lakshmanan, R.. \(2021, July 1\). IndigoZebra APT Hacking Campaign Targets the Afghan Government. Retrieved September 24, 2021.](#)
131. [CheckPoint Research. \(2021, July 1\). IndigoZebra APT continues to attack Central Asia with evolving tools. Retrieved September 24, 2021.](#)
132. [Artic Wolf Labs. \(2026, April 24\). Token Bingo: Don't Let Your Code be the Winner. Retrieved July 30, 2026.](#)
133. [Cimpanu, C.. \(2018, December 5\). Cyber-espionage group uses Chrome extension to infect victims. Retrieved August 26, 2019.](#)
134. [Tarakanov, D.. \(2013, September 11\). The "Kimsuky" Operation: A North Korean APT?. Retrieved August 13, 2019.](#)
135. [ThreatConnect. \(2020, September 28\). Kimsuky Phishing Operations Putting In Work. Retrieved October 30, 2020.](#)
136. [Kim, J. et al. \(2019, October\). KIMSUKY GROUP: TRACKING THE KING OF THE SPEAR PHISHING. Retrieved November 2, 2020.](#)
137. [Dahan, A. et al. \(2020, November 2\). Back to the Future: Inside the Kimsuky KGH Spyware Suite. Retrieved November 6, 2020.](#)
138. [An, J and Malhotra, A. \(2021, November 10\). North Korean attackers use malicious blogs to deliver malware to high-profile South Korean targets. Retrieved December 29, 2021.](#)
139. [KISA. \(2021\). Phishing Target Reconnaissance and Attack Resource Analysis Operation Muzabi. Retrieved March 8, 2024.](#)
140. [Den Iuzvyk, Tim Peck. \(2025, February 13\). Analyzing DEEP#DRIVE: North Korean Threat Actors Observed Exploiting Trusted Platforms for Targeted Attacks. Retrieved August 19, 2025.](#)
141. [Naumaan, S., et al. \(2025, April 17\). Around the World in 90 Days: State-Sponsored Actors Try ClickFix . Retrieved January 21, 2026.](#)
142. [Jazi, H. \(2021, February\). LazyScripter: From Empire to double RAT. Retrieved November 17, 2024.](#)
271. [Duncan, B. \(2020, July 24\). Evolution of Valak, from Its Beginnings to Mass Distribution. Retrieved August 31, 2020.](#)
272. [Duncan, B. \(2021, January 7\). TA551: Email Attack Campaign Switches from Valak to IcedID. Retrieved March 17, 2021.](#)
273. [Secureworks. \(n.d.\). GOLD CABIN Threat Profile. Retrieved March 17, 2021.](#)
274. [Trend Micro. \(2012\). The Taidoor Campaign. Retrieved November 12, 2014.](#)
275. [Livelli, K, et al. \(2018, November 12\). Operation Shaheen. Retrieved May 1, 2019.](#)
276. [Daniel Lugh, Jaromir Horejsi. \(2020, October 2\). Tonto Team - Exploring the TTPs of an advanced threat actor operating a large infrastructure. Retrieved October 17, 2021.](#)
277. [Huss, D. \(2016, March 1\). Operation Transparent Tribe. Retrieved June 8, 2016.](#)
278. [Dedola, G. \(2020, August 20\). Transparent Tribe: Evolution analysis, part 1. Retrieved September 2, 2021.](#)
279. [Malhotra, A. \(2021, March 2\). ObliqueRAT returns with new campaign using hijacked websites. Retrieved September 2, 2021.](#)
280. [Malhotra, A. et al. \(2021, May 13\). Transparent Tribe APT expands its Windows malware arsenal. Retrieved September 2, 2021.](#)
281. [Falcone, R. and Conant S. \(2016, March 25\). ProjectM: Link Found Between Pakistani Actor and Operation Transparent Tribe. Retrieved September 2, 2021.](#)
282. [Llimos, N., Pascual, C.. \(2019, February 12\). Trickbot Adds Remote Application Credential-Grabbing Capabilities to Its Repertoire. Retrieved March 12, 2019.](#)
283. [Ray, V. \(2016, November 22\). Tropic Trooper Targets Taiwanese Government and Fossil Fuel Provider With Poison Ivy. Retrieved November 9, 2018.](#)
284. [Alintanahin, K. \(2015\). Operation Tropic Trooper: Relying on Tried-and-Tested Flaws to Infiltrate Secret Keepers. Retrieved June 14, 2019.](#)
285. [Alexander, G., et al. \(2018, August 8\). Familiar Feeling: A Malware Campaign Targeting the Tibetan Diaspora Resurfaces. Retrieved June 17, 2019.](#)
286. [Moore, S. et al. \(2020, April 30\). Anomali Suspects that China-Backed APT Pirate Panda May Be Seeking Access to Vietnam Government Data Center. Retrieved May 19, 2020.](#)
287. [Chen, J.. \(2020, May 12\). Tropic Trooper's Back: USBferry Attack Targets Air gapped Environments. Retrieved May 20, 2020.](#)
288. [Harakhavik, Y. \(2020, February 3\). Warzone: Behind the enemy lines. Retrieved December 17, 2021.](#)
289. [Shinji Robert Arasawa, Joshua Aquino, Charles Steven Derion, Juhn Emmanuel Atanque, Francisrey Joshua Castillo, John Carlo Marquez, Henry Salcedo, John Rainier Navato, Arianne Dela Cruz, Raymart Yambot & Ian Kenefick. \(2024, January 9\). Black Basta-Affiliated Water Curupira's Pikabot Spam Campaign. Retrieved July 17, 2024.](#)
290. [Karim, T. \(2018, August\). TRAILS OF WINDSHIFT. Retrieved November 17, 2024.](#)
291. [Chad Anderson. \(2021, April 27\). Winter Vivern: A Look At Re-Crafted Government MalDocs Targeting Multiple Languages. Retrieved July 29, 2024.](#)
292. [Tom Hegel. \(2023, March 16\). Winter Vivern | Uncovering a Wave of Global Espionage. Retrieved July 29, 2024.](#)
293. [CERT-UA. \(2023, February 1\). UAC-0114 aka Winter Vivern to target Ukrainian and Polish GOV entities \(CERT-UA#5909\). Retrieved July 29, 2024.](#)

143. [Threat Intelligence Team. \(2021, August 23\). New variant of Konni malware used in campaign targetting Russia. Retrieved January 5, 2022.](#)

144. [Hawley, S. et al. \(2023, February 2\). Turla: A Galaxy of Opportunity. Retrieved May 15, 2023.](#)

145. [Conteras, T., Splunk Research Team. \(2025, September 25\). From Prompt to Payload: LAMEHUG's LLM-Driven Cyber Intrusion. Retrieved April 21, 2026.](#)

146. [Abrams, L. \(2024, April 30\). New Latrodectus malware attacks use Microsoft, Cloudflare themes. Retrieved September 13, 2024.](#)

147. [Sherstobitoff, R. \(2018, March 08\). Hidden Cobra Targets Turkish Financial Sector With New Bankshot Implant. Retrieved May 18, 2018.](#)

148. [Vyacheslav Kopeytsev and Seongsu Park. \(2021, February 25\). Lazarus targets defense industry with ThreatNeedle. Retrieved October 27, 2021.](#)

149. [Saini, A. and Hossein, J. \(2022, January 27\). North Korea's Lazarus APT leverages Windows Update client, GitHub in latest campaign. Retrieved January 27, 2022.](#)

150. [Pradhan, A. \(2022, February 8\). LolZarus: Lazarus Group Incorporating Lolbins into Campaigns. Retrieved March 22, 2022.](#)

151. [Axel F, Pierre T. \(2017, October 16\). Leviathan: Espionage actor spearphishes maritime and defense targets. Retrieved February 15, 2018.](#)

294. [Yamout, M. \(2021, November 29\). WIRTE's campaign in the Middle East 'living off the land' since at least 2019. Retrieved February 1, 2022.](#)

295. [John, E. and Carvey, H. \(2019, May 30\). Unraveling the Spiderweb: Timelining ATT&CK Artifacts Used by GRIM SPIDER. Retrieved May 12, 2020.](#)

296. [Brian Donohue, Katie Nickels, Paul Michaud, Adina Bodkins, Taylor Chapman, Tony Lambert, Jeff Felling, Kyle Rainey, Mike Haag, Matt Graeber, Aaron Didier.. \(2020, October 29\). A Bazar start: How one hospital thwarted a Ryuk ransomware outbreak. Retrieved October 30, 2020.](#)

297. [Shilko, J., et al. \(2021, October 7\). FIN12: The Prolific Ransomware Intrusion Threat Actor That Has Aggressively Pursued Healthcare Targets. Retrieved June 15, 2023.](#)

298. [MalwareBytes Threat Intelligence Team. \(2022, August 3\). Woody RAT: A new feature-rich malware spotted in the wild. Retrieved December 6, 2022.](#)

299. [Nart Villeneuve, Randi Eitzman, Sandor Nemes & Tyler Dean, Google Cloud. \(2017, October 5\). Significant FormBook Distribution Campaigns Impacting the U.S. and South Korea. Retrieved March 11, 2025.](#)

300. [Acronis. \(2021, November 26\). Trojan-as-a-service: From Formbook to XLoader. Retrieved March 11, 2025.](#)

301. [Microsoft. \(2020, October 13\). Anti-spoofing protection in EOP. Retrieved October 19, 2020.](#)

302. [Australian Cyber Security Centre. \(2012, December\). Mitigating Spoofed Emails Using Sender Policy Framework. Retrieved November 17, 2024.](#)